

Practical 6

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In [8]: # ===== Monoalphabetic Cipher =====
import random, string

key = list(string.ascii_lowercase)
random.shuffle(key)

def mono_encrypt(text):
    return ''.join(key[ord(c) - 97] if c.isalpha() else c for c in text.lower())

def mono_decrypt(text):
    return ''.join(chr(key.index(c) + 97) if c.isalpha() else c for c in text)

msg = "Hello, World!"
enc = mono_encrypt(msg)
dec = mono_decrypt(enc)

print("=== Monoalphabetic Cipher ===")
print(f"Original   : {msg}")
print(f"Encrypted   : {enc}")
print(f"Decrypted   : {dec}")
```

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=== Monoalphabetic Cipher ===
Original   : Hello, World!
Encrypted   : anqqo, xohqt!
Decrypted   : hello, world!
```

```
In [7]: # ===== Polyalphabetic Cipher (Vigenère) =====
def poly_encrypt(text, keyword):
    result, k = "", 0
    for c in text:
        if c.isalpha():
            shift = ord(keyword[k % len(keyword)]) - 97
            result += chr((ord(c.lower()) - 97 + shift) % 26 + 97)
            k += 1
        else:
            result += c
    return result

def poly_decrypt(text, keyword):
    result, k = "", 0
    for c in text:
        if c.isalpha():
            shift = ord(keyword[k % len(keyword)]) - 97
            result += chr((ord(c) - 97 - shift) % 26 + 97)
            k += 1
        else:
            result += c
    return result

msg = "Hello, World!"
```

```
enc = poly_encrypt(msg, "key")
dec = poly_decrypt(enc, "key")

print("=== Polyalphabetic Cipher (Vigenère) ===")
print(f"Original : {msg}")
print(f"Keyword   : key")
print(f"Encrypted  : {enc}")
print(f"Decrypted  : {dec}")
```

```
=== Polyalphabetic Cipher (Vigenère) ===
Original : Hello, World!
Keyword   : key
Encrypted : rijvs, uyvjn!
Decrypted : hello, world!
```